ROAD TRAFFIC ACCIDENTS (RTA) DATASET • TEAM 4

Road Traffic Accident Analysis

Graduation Project - DEPI Data Analysis Track

TEAM SIZE
5 members

DATASET
Road Traffic Accidents

BUSINESS REQUIREMENTS
40 questions

MAIN TOOL
Power BI

FINAL DEADLINE
4 Dec 2026

FINAL DEMO
1-2 minute demo video

Excel • Power Query • Power BI • DAX • GitHub

Project goal

Use Excel and Power BI to clean the road traffic accident data, answer 40 business questions, and build a clear interactive dashboard. The results will show which driver, vehicle, road, time and weather conditions go with serious and fatal accidents, and end in simple safety recommendations.

Tools we will use

- **Excel** — first look at the data, pivot-table checks and the data dictionary.
- **Power Query** — data cleaning and preparation inside Power BI.
- **Power BI Desktop** — data model, charts and dashboard.
- **DAX** — calculations and KPIs.
- **GitHub** — project files, documentation and final deliverables.

> **Note:** We will not use SQL or Python. Data cleaning and analysis will be done with Excel, Power Query, Power BI and DAX.

01 Project Plan and Deadlines

All project documents and final files go to the team GitHub repository.

Phase	What we will finish	DEPI deadline
1. Planning & Requirements	Project proposal, team roles, GitHub setup and the 40 business questions.	16 Oct 2026
2. Analysis & Data Design	Literature review, data dictionary, data model/schema, cleaning plan and dashboard page sketches.	6 Nov 2026
3. Implementation	Clean data, create DAX measures and build the Power BI dashboard with filters.	30 Nov 2026
4. Final Testing & Presentation	Test dashboard, finish report and slides, record demo video and prepare final delivery.	4 Dec 2026

Timeline

Work	Oct 2026	Nov 2026	Dec 2026
DEPI deadlines	16 Oct	6 Nov • 30 Nov	4 Dec
GitHub setup, confirm 40 questions	■		
Data dictionary, model, cleaning plan, sketches	■	■	
Clean data, DAX measures, start pages		■	
Finish dashboard, check every number		■	■
Report, video, final upload (slides in parallel)			■

Final deliverables checklist

- [ ] 40 business questions reviewed and approved.
- [ ] Dataset cleaned in Excel / Power Query and cleaning log saved.
- [ ] Data dictionary and data model/schema diagram completed.
- [ ] Power BI dashboard with charts, filters and the required DAX measures.
- [ ] DAX measures checked by a second team member.
- [ ] All dashboard pages and filters tested.
- [ ] Final report with key findings and simple recommendations.
- [ ] Final presentation slides.
- [ ] Demo video (1-2 minutes) recorded and included.
- [ ] All documents and project files uploaded to GitHub.

02 Team Members and Responsibilities

Team member	Main responsibilities
Retaj Etman Saber Elgallad Team Lead	Manage the project and deadlines; create and organize the GitHub repository; write the project proposal and risk plan; track tasks; create half of the required DAX measures; take part in cleaning the dataset; check that all files are uploaded.
Ibrahim Ahmed Adel Elbyaly	Clean and prepare the dataset using Excel and Power Query; keep the data cleaning log; help validate cleaned data; write and organize the final report.
Saif Eslam Fayeq Elsaedy	Prepare the data schema/model in Power BI; create the data dictionary; explain tables, columns, data types and relationships; check that the model relationships work correctly.
Habiba Talaat Abd El Azeem Farag	Plan the dashboard layout and visual design; choose suitable charts and filters; build the dashboard pages in Power BI; check that the dashboard is easy to read.
Ziad Hamdy Mostafa Abdelfattah	Create the remaining DAX measures; check calculations with the team; prepare the final presentation slides; present the main findings and recommendations.

Shared tasks

- Everyone reviews the 40 questions and confirms that the dataset has the needed columns.
- Everyone tests the dashboard and checks that totals match the cleaned data.
- The team lead checks progress, but each member is responsible for finishing their own tasks.
- Retaj and Ibrahim coordinate the final report; Ziad leads the slides; Habiba records the demo video with help from the team.

03 Dataset Overview

The project uses the Road Traffic Accidents (RTA) dataset. Each row is one accident, with 32 source columns about the driver, vehicle, road, environment, casualties and cause. There is no date, year, location or speed, only the time of day and the day of the week, and no outside data is added.

Dataset at a glance

12,316
Accidents (rows)

32
Source columns

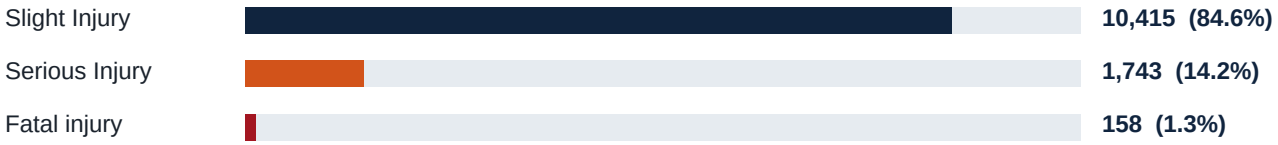
0
Duplicate rows

16
Columns with missing values

20,057
Missing cells

7,873
Rows with casualty details

Accident severity (imbalanced)



Source columns

Column names are kept exactly as in the source, including spelling mistakes (for example Road_allignment). In the casualty columns, 4,443 rows have "na" because no casualty details were recorded.

TIME AND DRIVER

Column	Data type	Meaning and notes
Time	Time	Time of day (hh:mm:ss).
Day_of_week	Text	Monday to Sunday.
Age_band_of_driver	Text	Under 18, 18-30, 31-50, Over 51, Unknown.
Sex_of_driver	Text	Male, Female, Unknown.
Educational_level	Text	Education level of the driver. 741 missing.
Vehicle_driver_relation	Text	Owner, Employee, Other. 579 missing.
Driving_experience	Text	Below 1yr to Above 10yr, or No Licence. 829 missing.

VEHICLE

Column	Data type	Meaning and notes
Type_of_vehicle	Text	Automobile, lorry, public bus, motorcycle, taxi, etc. 950 missing. Some labels have "?" instead of a dash, e.g. Lorry (41?100Q).
Owner_of_vehicle	Text	Owner, Governmental, Organization, Other. 482 missing.
Service_year_of_vehicle	Text	Below 1yr to Above 10yr. 3,928 missing (32%).
Defect_of_vehicle	Text	"No defect" or code 5 or 7 (not explained in the source). 4,427 missing (36%).
Vehicle_movement	Text	Going straight, overtaking, U-turn, reversing, etc. 308 missing.

ROAD

Column	Data type	Meaning and notes
Area_accident_occured	Text	Residential, office, school, market, rural, etc. 239 missing. One label is two values stuck together ("Rural village areasOffice areas").
Lanes_or_Medians	Text	One way, divided, undivided, etc. 385 missing.
Road_allignment	Text	Tangent, curve, steep grade, mountainous. 142 missing.
Types_of_Junction	Text	No junction, T / Y / O Shape, Crossing, Other. 887 missing.
Road_surface_type	Text	Asphalt, Earth, Gravel, etc. 172 missing.
Road_surface_conditions	Text	Dry, Wet or damp, Snow, Flood over 3cm.

ENVIRONMENT

Column	Data type	Meaning and notes
Light_conditions	Text	Daylight, or Darkness with lights lit, unlit or no lighting.
Weather_conditions	Text	Normal, Raining, Windy, Cloudy, Snow, Fog or mist, Other.

ACCIDENT

Column	Data type	Meaning and notes
Type_of_collision	Text	Vehicle with vehicle, rollover, with animals, with pedestrians, etc. 155 missing.
Number_of_vehicles_involved	Whole number	1 to 7.
Number_of_casualties	Whole number	1 to 8.
Cause_of_accident	Text	20 values (overspeed, overtaking, no priority, changing lane, etc.). Some labels look alike, e.g. "Overspeed" and "Driving at high speed".
Accident_severity	Text	Slight Injury, Serious Injury or Fatal injury.

CASUALTY (DESCRIPTION ONLY)

Column	Data type	Meaning and notes
Casualty_class	Text	Driver or rider, Passenger, Pedestrian.
Sex_of_casualty	Text	Male or Female.
Age_band_of_casualty	Text	Age group. Code "5" appears in 244 rows (not explained in the source).
Casualty_severity	Text	Code 1, 2 or 3 (not explained in the source).
Work_of_casuality	Text	Driver, employee, student, etc. 3,198 missing.
Fitness_of_casuality	Text	Normal, Deaf, Blind, Other. 2,635 missing. 19 rows say "NormalNormal" (typing error).
Pedestrian_movement	Text	Movement of the pedestrian, or "Not a Pedestrian". Labels contain a typing error ("statioNot a Pedestrianry" instead of "stationary").

Columns added in Power BI

Column	Data type	Meaning
Hour	Whole number	Hour of the day (0-23) taken from Time.
Day Number	Whole number	Monday = 1 to Sunday = 7, used to sort the days correctly.
Is Weekend	Text	Weekend (Saturday, Sunday) or Weekday.
Serious or Fatal	Text	Yes if Accident_severity is Serious Injury or Fatal injury, otherwise No.
Light Group	Text	Daylight or Darkness (the three darkness values grouped).
At Junction	Text	Yes for any junction type except "No junction", otherwise No.
Has Defect	Text	Yes if Defect_of_vehicle has a defect code, No if "No defect", Unknown if missing.

Definitions used in the analysis

- **Accident:** one row of the dataset.
- **Serious or fatal accident:** Accident_severity is Serious Injury or Fatal injury.
- **Percentage of serious or fatal accidents:** serious or fatal accidents divided by all accidents in the group.
- **Weekend:** Saturday and Sunday (team definition; results change with the definition).

04 The 40 Business Requirements

The dashboard should answer these questions. Each one can be answered from the available columns. "Serious or fatal" means Serious Injury or Fatal injury.

A. Driver

1	Which driver age band is involved in the most accidents?
2	Which driver age band has the highest percentage of serious or fatal accidents?
3	How many accidents involve male drivers and how many involve female drivers?
4	Which driving experience group has the most accidents?
5	Which driving experience group has the highest percentage of serious or fatal accidents?
6	Which education level has the most accidents?
7	How many accidents involve drivers with no licence?
8	What share of accidents comes from each driver-vehicle relation (owner, employee, other)?

B. Vehicle

9	Which vehicle type is involved in the most accidents?
10	Which vehicle type has the highest percentage of serious or fatal accidents (types with at least 30 accidents)?
11	Which vehicle owner type has the most accidents?
12	Which vehicle service-year group has the most accidents?
13	How many accidents involve a vehicle with a recorded defect?
14	Which vehicle movement is the most common in accidents?
15	What is the average number of vehicles involved per accident?
16	What percentage of accidents involve more than two vehicles?

C. Road and Junction

17	Which junction type has the most accidents?
18	What percentage of accidents happens at a junction (any type except "No junction")?
19	Which lane or median type has the most accidents?
20	Which road alignment has the most accidents?
21	Which road surface type has the most accidents?
22	Which area type has the most accidents?
23	Which type of collision is the most common?
24	Which type of collision has the highest percentage of serious or fatal accidents?

D. Environment and Time

25	At which hour of the day do the most accidents happen?
26	Which day of the week has the most accidents?
27	Which day of the week has the highest percentage of serious or fatal accidents?
28	What percentage of accidents happens in daylight and what percentage in darkness?
29	Which light condition has the highest percentage of serious or fatal accidents?
30	Which weather condition has the most accidents?
31	How many accidents happen on each road surface condition (dry, wet, snow, flood)?
32	What percentage of accidents happens on weekends (Saturday and Sunday)?

E. Causes and Casualties

33	Which cause of accident is the most frequent?
34	Which five causes have the highest percentage of serious or fatal accidents (causes with at least 30 accidents)?
35	What share of all accidents does each of the top five causes represent?
36	What is the average number of casualties per accident?
37	What percentage of accidents has more than one casualty?
38	What share of casualties are drivers or riders, passengers and pedestrians?
39	Which casualty age band has the most casualties?
40	How many accidents are Slight, Serious and Fatal, and what is the percentage of each?

05 How We Will Build the Project

We work in five steps, from the raw file to the tested dashboard.

Step 1 — Prepare the data

- Use Excel for a first check, then Power Query to check missing values, duplicates, wrong labels and data types.
- Keep a short cleaning log that lists what was changed and why. It records: spaces trimmed in text columns; 0 duplicate rows found; missing and "na" values marked "Unknown" (not deleted, because no outside data is allowed); typing errors fixed (Pedestrian_movement, "NormalNormal", the merged area label); and the helper columns added (see Dataset Overview).
- Unexplained codes (Defect_of_vehicle, Casualty_severity, Age_band_of_casualty) are kept as codes and described in the report.

Step 2 — Build the data model

- Write each column name, meaning and data type in the data dictionary (data_dictionary_RTA.xlsx).
- Create a clear model in Power BI: one cleaned Accidents table plus small helper tables for sorting (day of week, severity), and check that their relationships work.

Step 3 — Create DAX measures

- Reusable measures: Total Accidents, Serious or Fatal Accidents, Fatal Accidents, Serious Share %, Share of Total %, Average Casualties, Average Vehicles Involved, and rankings (Top N).
- Retaj creates about half of the measures and Ziad creates the rest. They review each other's results.

Step 4 — Build the dashboard

- Use clear charts, cards, slicers and titles, with a simple and consistent layout.
- Pages: Overview, Driver, Vehicle, Road and Junction, Environment and Time, Causes and Casualties.
- Slicers: severity, day of week, hour, weather, light, vehicle type, age band.

Step 5 — Test and finish

- Check key totals and every measure against the cleaned data (Excel pivot tables are the reference).
- Test slicers, page navigation and chart interactions; hide or flag groups with fewer than 30 accidents.
- Write the report and recommendations, prepare the slides, and record the demo video.

Suggested dashboard KPIs

Total Accidents
All accidents (12,316 before any filter)

Serious or Fatal Share
Serious plus fatal accidents divided by all accidents

Fatal Accidents
Fatal accidents (158 before any filter)

Average Casualties
Average number of casualties per accident

Top Cause
Most frequent cause in the current filter

Peak Hour
Hour with the most accidents in the current filter

06 Risk Assessment and Project KPIs

The main risks are small groups (only 158 fatal accidents), missing data and wrong interpretation of the results. Each risk has a simple plan to reduce it.

Risk assessment and mitigation plan

Risk	Likelihood	Impact	Mitigation
Dashboard numbers do not match the cleaned data	Medium	High	Compare every key number with an Excel pivot table before delivery; each member tests the totals.
Small groups give misleading rates (158 fatal accidents, rare vehicle types)	High	High	Show counts next to rates; mark groups with fewer than 30 accidents as too small to compare.
Missing values (Defect_of_vehicle 36%, Service_year_of_vehicle 32%) bias the results	High	Medium	Keep "Unknown" as its own category and state it in the report; do not guess values.
Wrong conclusions: association shown as cause, or casualty columns used as risk factors	Medium	High	Use words like "is linked to"; keep casualty columns for description only; write the definitions in the report.
Percentage and Top N visuals give wrong results when filters are used	Medium	High	Test every slicer; build the measures so rates always divide by the accidents in the current filter.
Power BI file is lost or members overwrite each other's work	Medium	High	Upload a version to GitHub after each phase; only one member edits the model at a time.
The CSV path is tied to one computer, so Refresh fails for other members	High	Medium	Use a file-path parameter, or keep the CSV inside the repository.
Not enough time before the 4 Dec 2026 deadline	Medium	High	Prepare slides and demo video in parallel with the dashboard; assign tasks clearly.
A team member is late with a task	Medium	High	Weekly progress check by the team lead; members help each other when someone falls behind.
Limited data: no date, location, speed or driver age in years	Fixed	Medium	State the scope clearly; keep recommendations about conditions, time of day and causes only.

Project KPIs (measures of project success)

Project KPI	Target	How we check it
Business questions answered in the dashboard	40 of 40	Tick each question against a chart, card or table
Dashboard numbers matching the cleaned data	100%	Compare dashboard cards with Excel pivot tables
DAX measures checked by another team member	All measures	A second member reviews each measure
Dashboard pages tested (slicers, navigation, interactions)	All pages	Test every slicer and page before delivery
Deliverables uploaded to GitHub	All, before 4 Dec 2026	Team lead checks the repository before each deadline
Phase deadlines met (16 Oct, 6 Nov, 30 Nov, 4 Dec)	4 of 4	Weekly progress check by the team lead

07 Problem Statement, Objectives and Scope

Problem statement

The RTA data has 12,316 accident records, but it is raw: 16 columns have missing values, some labels contain typing errors, and 84.6% of accidents are slight, so serious and fatal accidents are easy to miss. As it comes, it is hard to tell which drivers, vehicles, roads, times and weather conditions go with serious accidents, and which causes are most common.

Project objectives

Clean this data and use it to answer 40 business questions about drivers, vehicles, roads, environment and time, and accident causes and casualties. Build an interactive Power BI dashboard with six pages (Overview, Driver, Vehicle, Road and Junction, Environment and Time, Causes and Casualties), then write a report with the main findings and simple safety recommendations, plus the slides and a demo video.

Project scope

We only use the RTA dataset as it is, with no outside data. Because it has no date, location or speed, there is no yearly trend, map or speed analysis. We describe links between conditions and severity; we do not claim that one condition causes another. Everything is done with Excel, Power Query, Power BI and DAX (no SQL or Python).